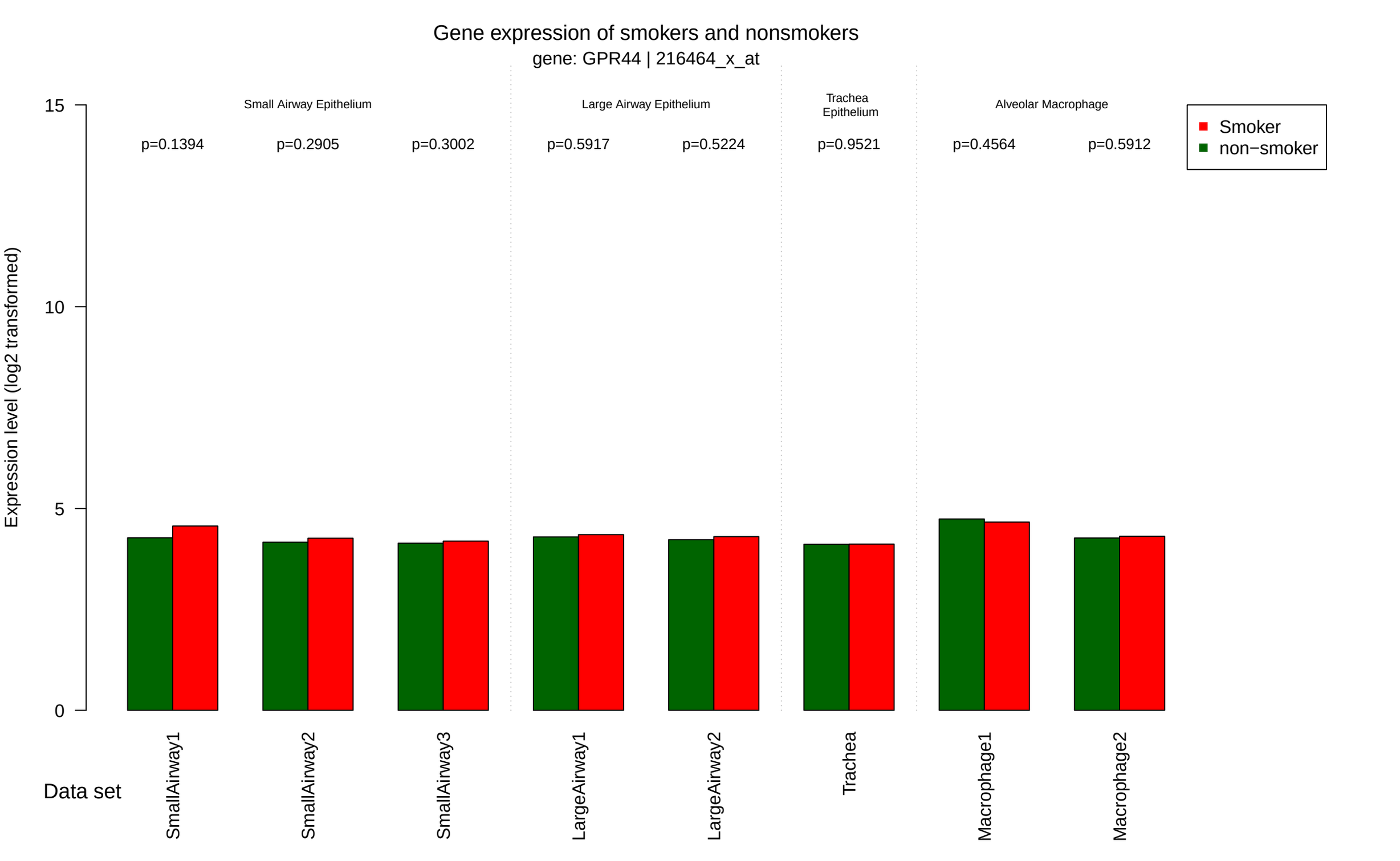
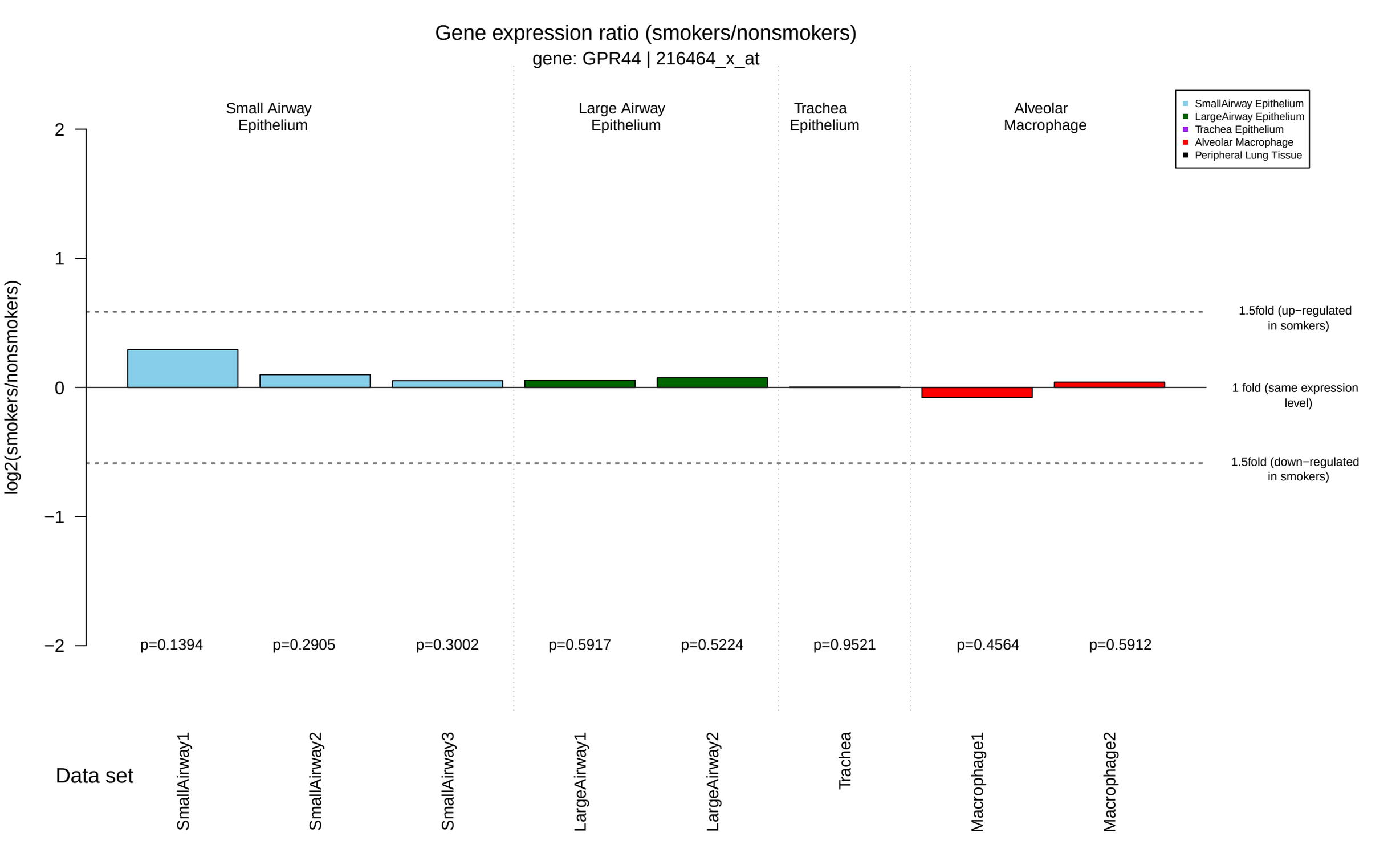
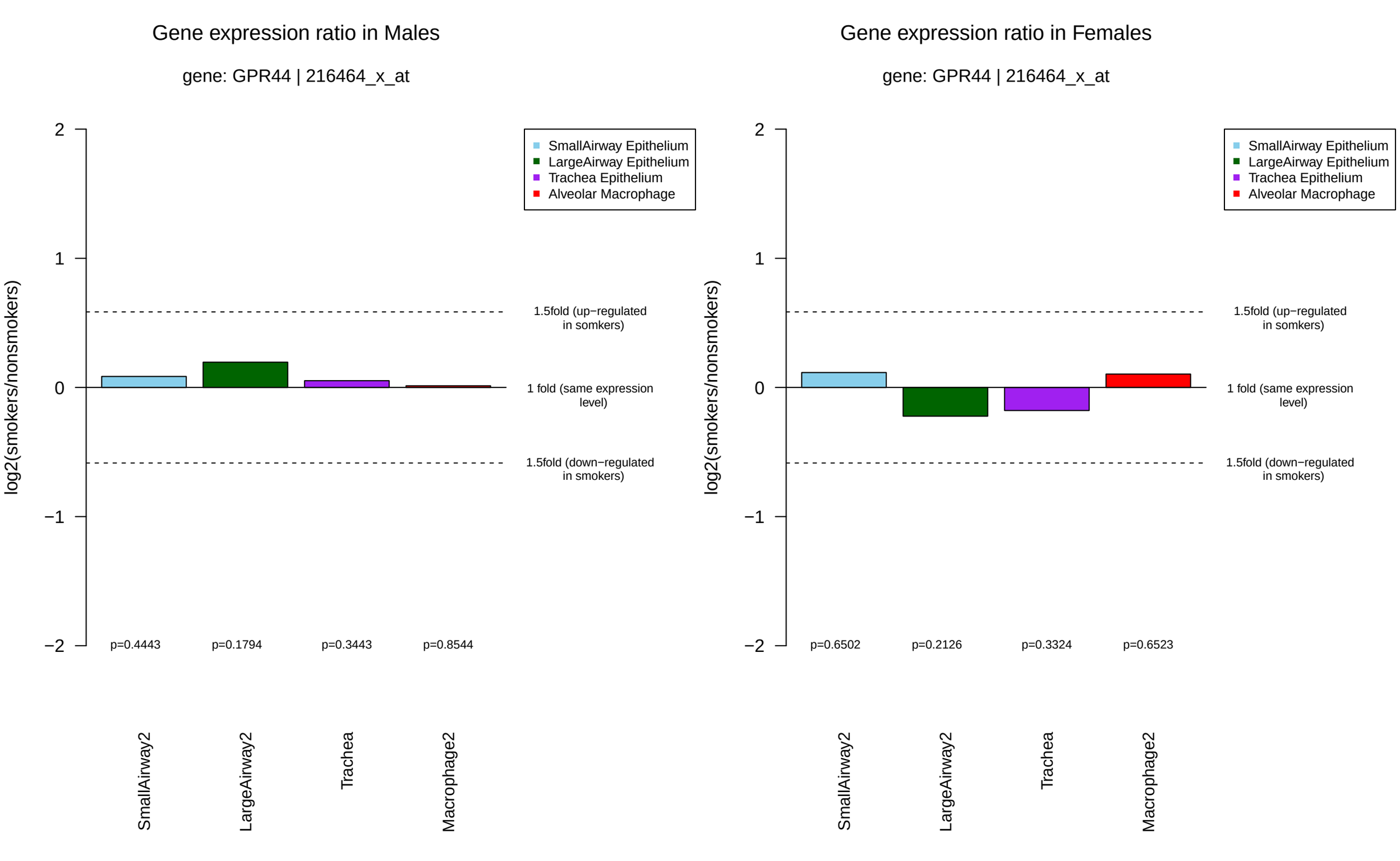
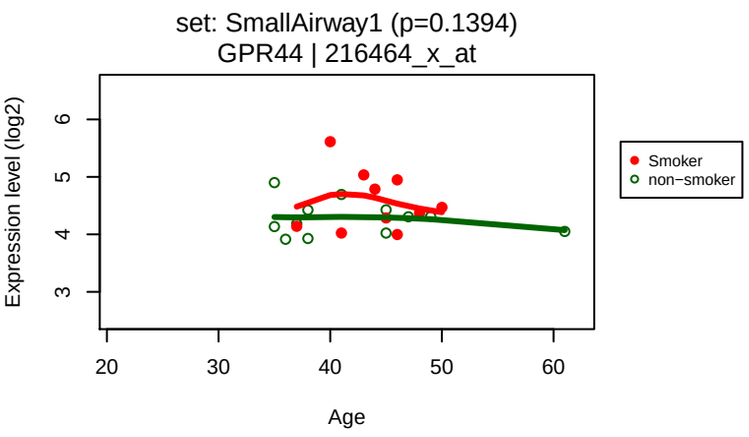
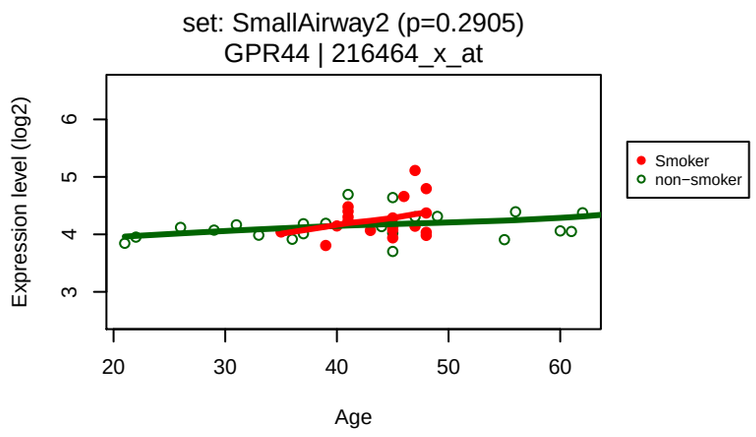
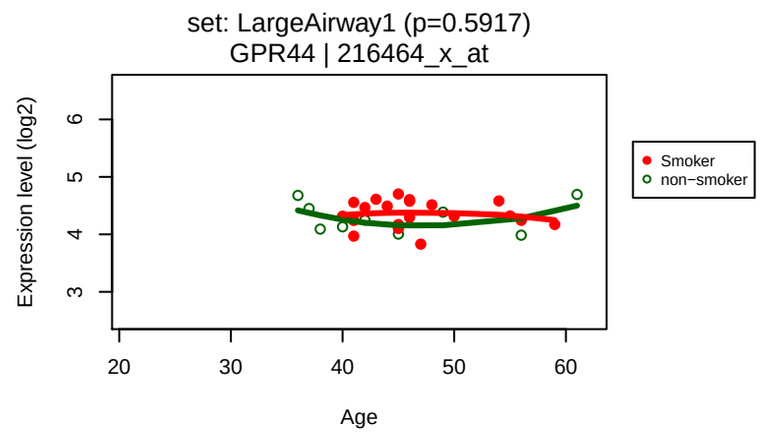

Gene expression of smokers and nonsmokers
gene: GPR44 | 216464_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1394
p=0.2905
p=0.3002
p=0.5917
p=0.5224
p=0.9521
p=0.4564
p=0.5912
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

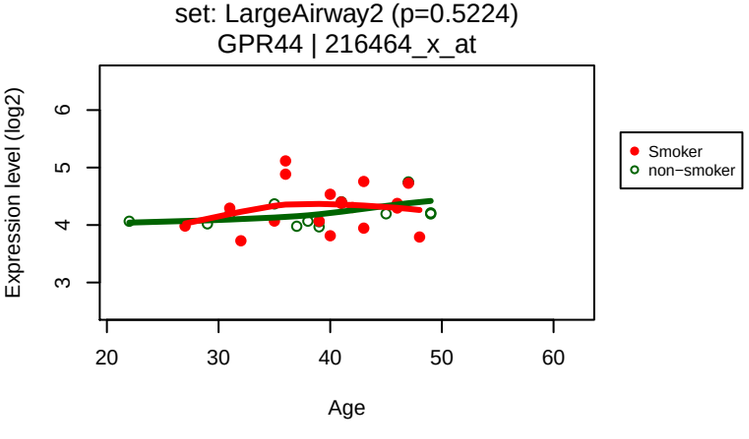

Gene expression ratio (smokers/nonsmokers)
gene: GPR44 | 216464_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1394
p=0.2905
p=0.3002
p=0.5917
p=0.5224
p=0.9521
p=0.4564
p=0.5912
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

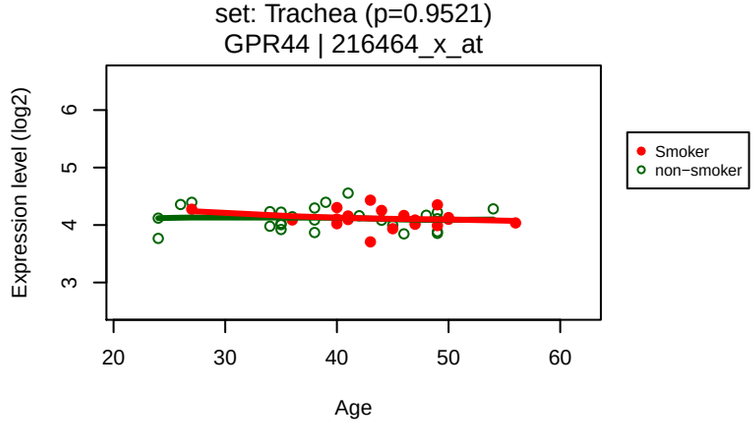

Gene expression ratio in Males
gene: GPR44 | 216464_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4443
p=0.1794
p=0.3443
p=0.8544
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR44 | 216464_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6502
p=0.2126
p=0.3324
p=0.6523
SmallAirway2
LargeAirway2
Trachea
Macrophage2

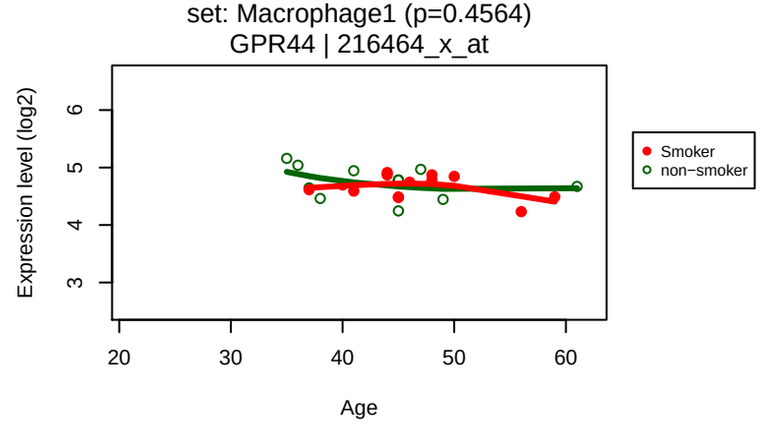

set: SmallAirway1 (p=0.1394)
GPR44 | 216464_x_at
Expression level (log2)
Age
Smoker
non−smoker

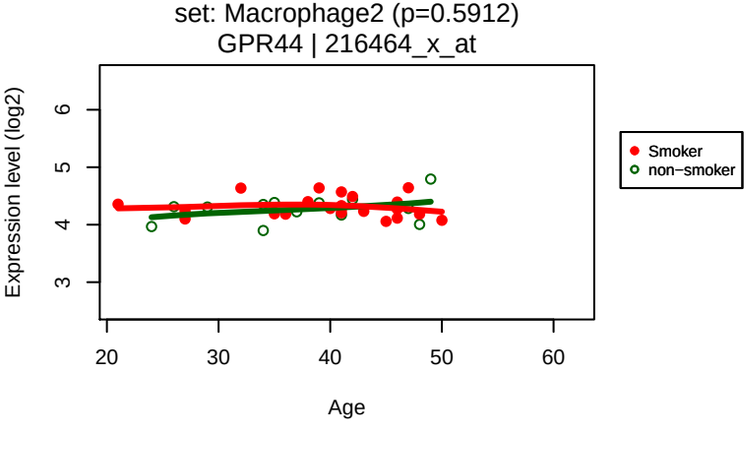

set: SmallAirway2 (p=0.2905)
GPR44 | 216464_x_at
Expression level (log2)
Age
Smoker
non−smoker

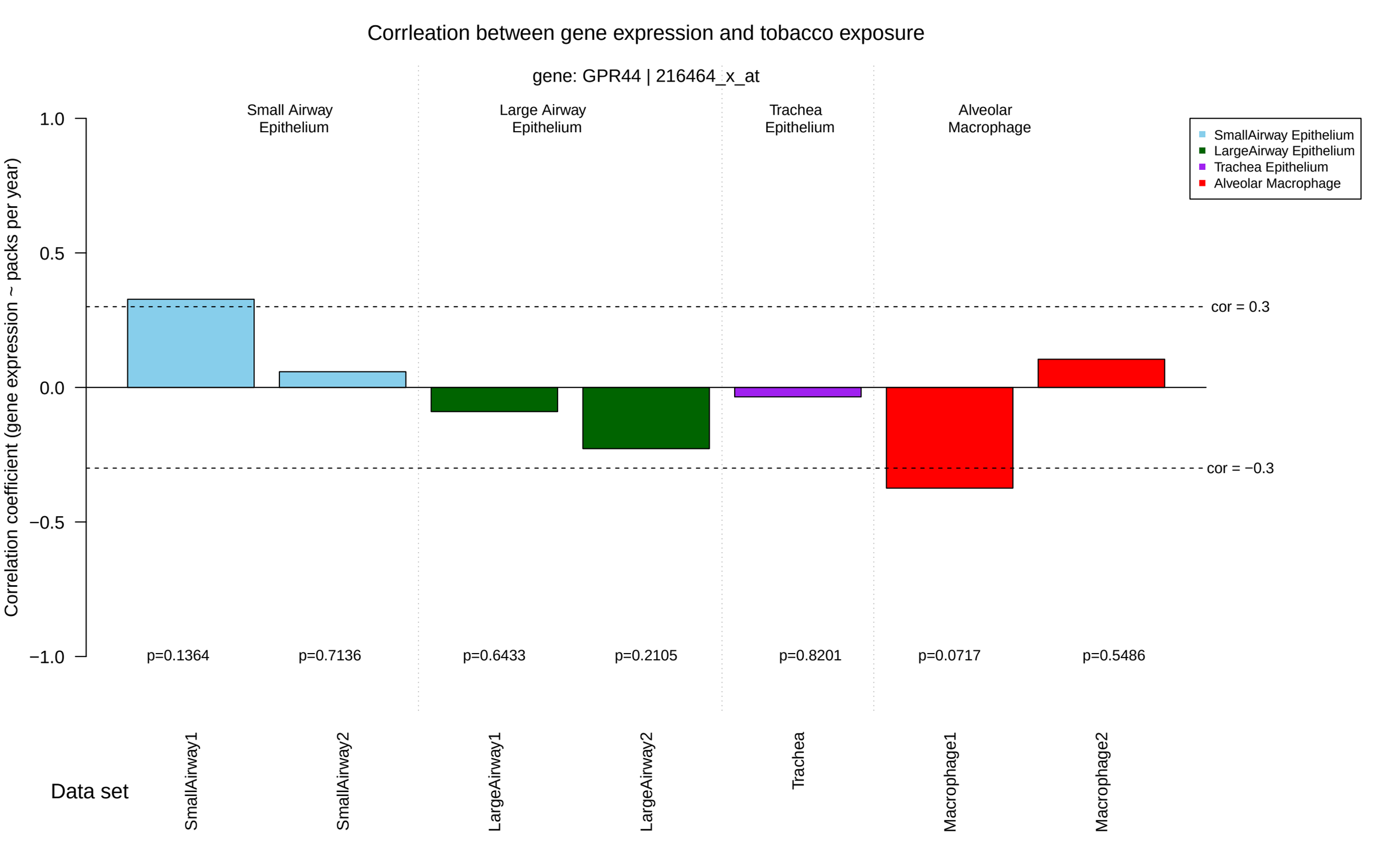

set: LargeAirway1 (p=0.5917)
GPR44 | 216464_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5224)
GPR44 | 216464_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9521)
GPR44 | 216464_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4564)
GPR44 | 216464_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5912)
GPR44 | 216464_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR44 | 216464_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1364
p=0.7136
p=0.6433
p=0.2105
p=0.8201
p=0.0717
p=0.5486
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2